

MODULE 2.5

Hardening Wireless and Mobile Devices

Network Defense | Cisco Networking Academy

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Topics Covered:

- 2.5.1 • Wireless Security Overview & Threats
- 2.5.2 • Wireless Security Protocols — WEP, WPA, WPA2, WPA3
- 2.5.3 • Hardening a Wireless LAN
- 2.5.4 • Mobile Device Security & MDM
- 2.5.5 • Bluetooth Security

2.5.1 Wireless Security Overview & Threats

Why wireless networks require special attention when hardening

Why Wireless is More Vulnerable

- Wireless networks transmit data over radio waves — unlike wired networks, the signal travels through the air and can be intercepted by anyone within range.
- There are no physical boundaries. An attacker doesn't need to plug into the network — they just need to be close enough to receive the signal.
- For @Apollo, unsecured wireless access points are an open door into the entire network.

Common Wireless Threats

- Eavesdropping — Intercepting wireless traffic to capture usernames, passwords and sensitive data.
- Rogue Access Points — Attacker sets up a fake Wi-Fi hotspot that mimics a legitimate one. Users connect unknowingly and all their traffic is captured.
- Evil Twin Attack — A rogue AP with the same SSID as the real network, tricking devices into connecting to it instead.
- Denial of Service (DoS) — Flooding the wireless channel with traffic to prevent legitimate users from connecting.
- Unauthorized Access — Connecting to an unsecured or weakly secured wireless network to gain access to the internal network.

2.5.2 Wireless Security Protocols

The evolution from WEP to WPA3 — and why the version matters

Wireless Security Protocols — Oldest to Newest

- WEP — Wired Equivalent Privacy (Do NOT use)

The original wireless security protocol. Uses RC4 encryption with a static key. Severely flawed — an attacker can crack a WEP key in minutes. Completely deprecated.

- WPA — Wi-Fi Protected Access (Avoid if possible)

Introduced TKIP (Temporal Key Integrity Protocol) to replace the broken WEP keys. A significant improvement, but TKIP has since been found to have weaknesses.

- WPA2 — Wi-Fi Protected Access 2 (Current standard)

Replaced TKIP with AES-CCMP encryption — much stronger and considered secure. Available in Personal (PSK) and Enterprise (802.1X) modes. Use this as the minimum.

- WPA3 — Wi-Fi Protected Access 3 (Latest — Recommended)

Uses SAE (Simultaneous Authentication of Equals) for stronger handshake protection. Protects against offline dictionary attacks and provides forward secrecy. Use wherever possible.

2.5.3 Hardening a Wireless LAN

Practical steps to secure @Apollo's wireless network

Wireless LAN Hardening Checklist

- Change default SSID — Do not use the manufacturer's default network name, which reveals the router brand and model to attackers.
- Change default admin credentials — Default usernames and passwords are publicly known. Change them immediately on every access point.
- Use WPA2 or WPA3 — Never use WEP or open (unsecured) wireless. Always require a strong encryption standard.
- Disable SSID broadcasting — Hiding the network name makes it harder for casual attackers to discover the network exists.
- Enable MAC address filtering — Only allow known, approved device MAC addresses to connect to the wireless network.
- Position access points carefully — Place APs away from exterior walls and windows to minimise signal leakage outside the building.
- Disable remote management — The router admin panel should only be accessible from the internal wired network, not remotely over the internet.
- Keep firmware updated — Router and AP firmware updates patch known security vulnerabilities.
- Use a separate guest network — Visitors should connect to an isolated network with no access to internal resources.

2.5.4 Mobile Device Security & MDM

Managing and securing smartphones, tablets and laptops in the workplace

The Mobile Security Challenge

- Mobile devices are portable, always connected, and often contain sensitive organisational data.
- BYOD (Bring Your Own Device) policies allow employees to use personal devices for work — creating a significant security challenge for organisations like @Apollo.
- Lost or stolen devices, unsecured apps and unencrypted data are the most common risks.

Mobile Device Management (MDM)

- MDM software allows an organisation to manage and secure all mobile devices connected to the corporate network from a central location.
- Remote Wipe — If a device is lost or stolen, MDM can erase all data on it remotely.
- Screen Lock & PIN Enforcement — MDM enforces strong lock screens and passwords on all managed devices.
- App Management — Only approved applications can be installed. Unapproved or malicious apps can be remotely blocked or removed.
- Data Encryption — MDM enforces full device encryption so that data is unreadable if the device falls into the wrong hands.
- Location Tracking — Allows the organisation to locate a missing device.
- BYOD Policy — Clear rules must define which personal devices can access corporate resources, and under what conditions.

2.5.5 Bluetooth Security

Threats and countermeasures for Bluetooth-enabled devices

Bluetooth Threats

- Bluejacking — Sending unsolicited messages to a Bluetooth-enabled device. Mostly harmless but can be used to send phishing links or annoying spam.
- Bluesnarfing — Unauthorised access to information on a Bluetooth device including contacts, messages, emails and calendar data. This is illegal and a serious privacy violation.
- Bluebugging — An attacker gains full control of a target's Bluetooth device, allowing them to make calls, send messages and access all data without the owner knowing.
- Man-in-the-Middle (MITM) — Intercepting Bluetooth communication between two devices to eavesdrop or alter data being exchanged.

Bluetooth Hardening Countermeasures

- Turn off Bluetooth when it is not actively being used — a device that is not discoverable cannot be targeted.
- Never accept pairing requests from unknown or untrusted devices.
- Set devices to 'non-discoverable' mode in public places.
- Use strong Bluetooth PINs for pairing — avoid simple codes like 0000 or 1234.
- Keep device software and Bluetooth firmware updated to patch known vulnerabilities.

2.5 Summary — Hardening Wireless and Mobile Devices

Key concepts from this module

Wireless Security

- WEP — Broken, never use
- WPA — Weak, avoid if possible
- WPA2 — Current minimum standard (AES)
- WPA3 — Latest, strongest (SAE)

- Change default SSID and admin credentials
- Disable SSID broadcast and remote management
- Enable MAC filtering and use a guest network
- Position APs to limit signal leakage
- Keep firmware patched and updated

Mobile & Bluetooth Security

- Use MDM to centrally manage all mobile devices
- Enforce screen locks, strong PINs and encryption
- Enable remote wipe for lost or stolen devices
- Control apps through MDM — approved apps only
- Implement a clear BYOD policy

- Bluejacking — unsolicited Bluetooth messages
- Bluesnarfing — unauthorised data access via Bluetooth
- Bluebugging — full device takeover via Bluetooth
- Turn off Bluetooth when not in use and never pair with unknown devices